

What do we learn:-

- Bits, Qubits, Q-states
- changes of measurements
- state representation

- Every PC works with info.
- classical use bits
- quantum use qubits

Bits	Qubits
smallest unit of info in PC	Basic unit in Q-computer
have one value at a time	not restricted to one state
0 or 1	0, 1, combined
M: Switch-OFF/ON	M: spinning coin
Multiple Bits do process large info (img, audio)	explore the large solution spaces
Ex: 0001 - 1 0010 - 2 0100 - 4 1000 - 8	Not Imaginary ↓ Superconducting circuits trapped ions photons

Quantum State:-

- Describe the complete condition of a qubit.
- tells about a qubit before measurement
- Derive the probability
- state representation

↳ $|0\rangle, |1\rangle$

↳ Ket Notation

→



Superposition:- 50% - 50-50 chances

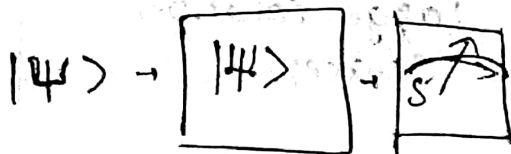
- the quantum state stores these probs until measurement occurs



Measurement:-

Before: JUST combination of probabilities

after:- Either 0/1



How it works:-

